

HORNSBY GIRLS' HIGH SCHOOL



2008 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question
- Begin each question on a fresh sheet of paper

Total marks (120)

- Attempt Questions 1 – 8
- All questions are of equal value

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Attempt Questions 1 – 8

All Questions are of equal value

Begin each question on a NEW SHEET of paper, writing your student number and question number at the top of the page. Extra paper is available.

Question 1 (15 marks) Use a SEPARATE sheet of paper.	Marks
(a) Find $\int \frac{dx}{x^2 - 4x + 40}$	2
(b) Evaluate $\int_0^2 x^3 e^{x^2} dx$.	3
(c) Find $\int \sin^3 x dx$	2
(d) Evaluate $\int_0^1 \frac{x}{\sqrt{4-x}} dx$	3
(e) (i) Find the real numbers a , b and c such that $\frac{3x^2 + 2x + 11}{(x^2 + 3)(1-x)} \equiv \frac{ax + b}{x^2 + 3} + \frac{c}{1-x}.$	3
(ii) Hence find $\int \frac{3x^2 + 2x + 11}{(x^2 + 3)(1-x)} dx$.	2

Question 2 (15 marks) Use a SEPARATE sheet of paper.

Marks

(a) Given z is a complex number such that $z = 1 + i$

(i) Write z in mod-arg form

2

(ii) Evaluate z^{12}

2

(b) If $P(z) = z^4 - 30z^2 + 289$

(i) Show that $z = 4 + i$ is a zero of $P(z)$

2

(ii) Find all zeros of $P(z)$ over the complex field

5

(c) $P(z)$ is a point on the argand diagram such that

4

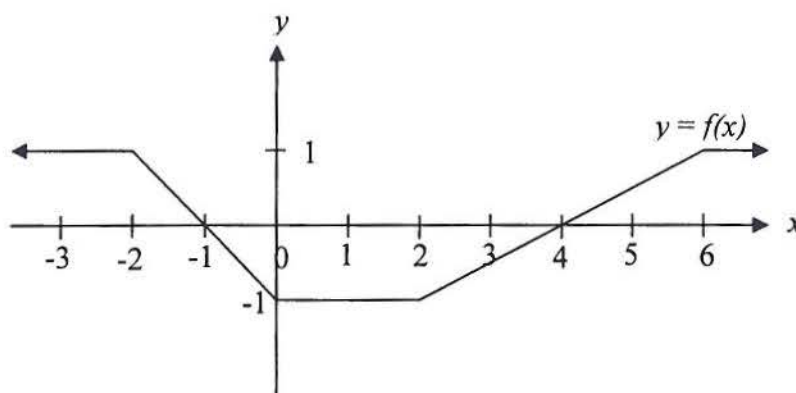
$$\arg \frac{z-i}{z+2} = \frac{\pi}{2}$$

Draw and describe the locus of $P(z)$.

Question 3 (15 marks) Use a SEPARATE sheet of paper.

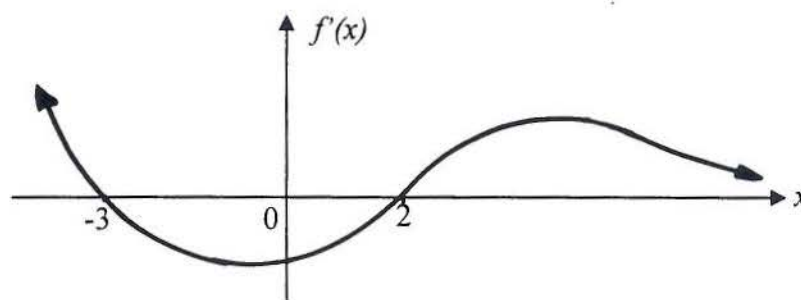
Marks

- (a) The diagram below is a sketch of the function $y = f(x)$



On separate diagrams sketch

- (i) $y = |f(x)|$ 2
- (ii) $y = f(|x|)$ 1
- (b) The graph below represents the derivative $f'(x)$ of a certain function $f(x)$. 3
- Given that $f'(x) \rightarrow 0$ as $x \rightarrow \infty$, $f(0) = 0$ and $f(1) < 0$, sketch the graph of $f(x)$, noting the behaviour as $x \rightarrow \infty$.



- (c) (i) Sketch the curve $y = \frac{x^3 + 4}{x^2}$, showing any stationary points and asymptotic behaviour. 2
- (ii) Hence or otherwise, deduce the values of k , for which the equation $x^3 - kx^2 + 4 = 0$ may have one real root. 1
- (d) (i) If $x = a$ is a multiple root of the polynomial equation $P(x)$ such that $P(x) = 0$, prove that $P'(a) = 0$. 3
- (ii) Find all roots of $P(x) = 16x^3 - 12x^2 + 1$ given that two of the roots are equal. 3

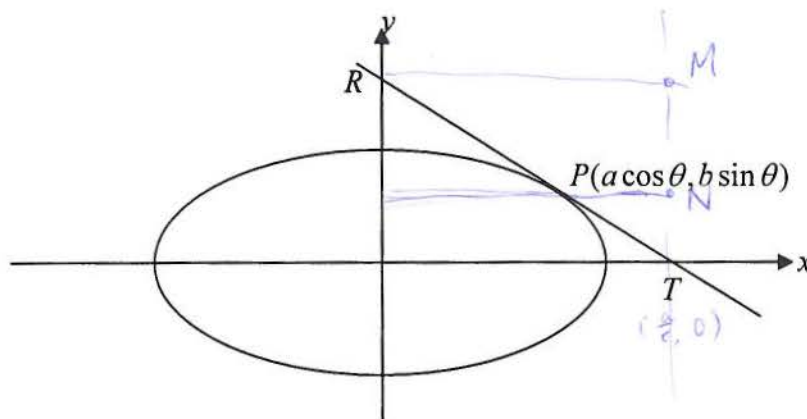
Question 4 (15 marks) Use a SEPARATE sheet of paper.**Marks**

- (a) An ellipse has parametric equations $x = \sqrt{2} \cos \theta$ and $y = 3 \sin \theta$.

2

Find the Cartesian equation and the eccentricity of the ellipse.

- (b)



The ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ shown in the diagram above has a tangent at the point $P(a \cos \theta, b \sin \theta)$. The tangent cuts the x -axis at T and the y -axis at R .

- (i) Show that the equation of the tangent at the point P is

2

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$$

- (ii) If T is the point of intersection between the tangent at point P and one of the directrices of the ellipse, show that $\cos \theta = e$.

3

- (iii) Hence find the angle that the focal chord through P makes with the x -axis.

1

- (iv) Using similar triangles or otherwise, show that $RP = e^2 RT$.

3

- (c) The area between the curve $y = \ln(x+1)$ and the x -axis, between $x = 0$ and $x = 1$ is rotated about the y -axis.

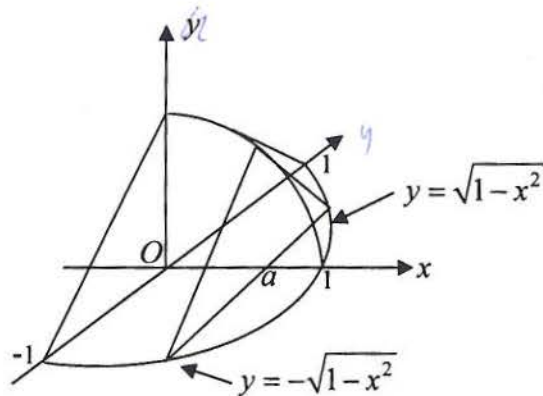
4

Find the volume of the solid of revolution formed using the method of cylindrical shells.

Question 5 (15 marks) Use a SEPARATE sheet of paper.**Marks**

- (a) (i) Write down the value of $\int_a^a \sqrt{a^2 - x^2} dx$. 1
- (ii) Explain why $\int_a^a x\sqrt{a^2 - x^2} dx$ is equal to zero. 1

(b)



The base of a solid is the semi-circular region in the x - y plane with the straight edge running from the point $(0,-1)$ to the point $(0,1)$ and the point $(1,0)$ on the curved edge of the semicircle.

Each cross-section perpendicular to the x -axis is an isosceles triangle with each of the two equal side lengths three quarters the length of the third side.

- (i) Show that the area of the triangular cross-section at $x = a$ is $\frac{\sqrt{5}}{2}(1 - a^2)$. 2
- (ii) Hence find the volume of the solid. 2
- (c) The point $T\left(ct, \frac{c}{t}\right)$ lies on the hyperbola $xy = c^2$. The tangent at T meets the x -axis at P and the y -axis at Q . The normal at T meets the line $y=x$ at R .
- (i) Prove that the tangent at T has equation $x + t^2 y = 2ct$. 2
- (ii) Find the coordinates of P and Q . 2
- (iii) Write down the equation of the normal at T . 1
- (iv) Show that the x coordinate of R is $x = \frac{c}{t}(t^2 + 1)$. 2
- (v) Prove that $\triangle PQR$ is isosceles. 2

Question 6 (15 marks) Use a SEPARATE sheet of paper.**Marks**

- (a) The equation $x^3 + 2x - 1 = 0$ has roots α, β, γ .

Find a polynomial equation in x whose roots are:

(i) $-\alpha, -\beta, -\gamma$

1

(ii) $\alpha^2, \beta^2, \gamma^2$

2

(iii) $\pm\alpha, \pm\beta, \pm\gamma$

2

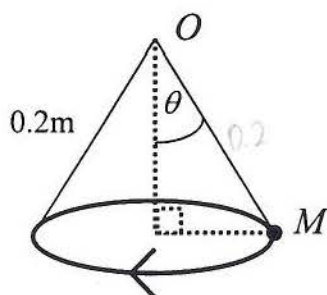
- (b) Find a and b if $(1+i)$ is a root of $x^2 + (a+2i)x + 5 - ib = 0$

3

- (c) A body M , of mass 650g , is fixed to point O by a light wire 0.2m long.

The body rotates in a horizontal plane at 72 revolutions per minute.

Taking $g = 10\text{m/s}^2$,



(i) Prove that $\tan \theta = \frac{72\pi^2 \sin \theta}{625}$.

3

- (ii) Find θ to the nearest minute.
rad.?

2

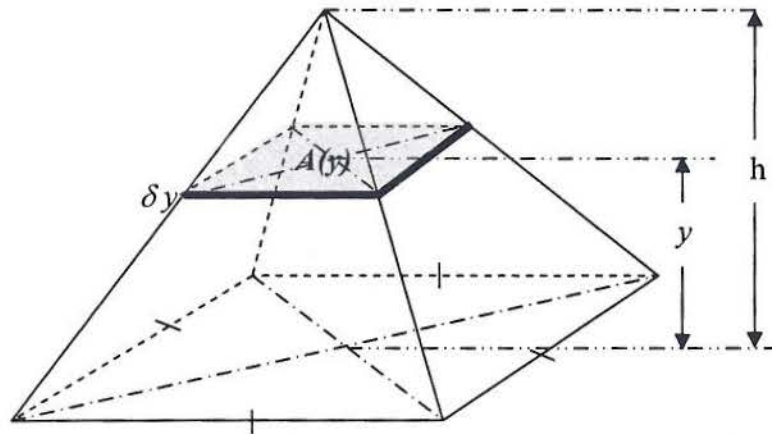
- (iii) The mass of the body is to be doubled but the speed of rotation is to remain the same. What will happen to the value of θ ?

2

Question 7 (15 marks) Use a SEPARATE sheet of paper.

Marks

- (a) The great pyramid of Cheops at Giza in Egypt is approximately 150 m high and its base is a square with an area of approximately 5 hectares.



- (i) Show that the area of the cross section $A(y)$, at y is given by

1

$$A(y) = (5 \times 10^4) \times \left(\frac{h-y}{h} \right)^2$$

- check* (ii) Find the volume of the pyramid by using the slicing technique.

4

- (b) A particle of mass 1 kg is projected vertically upwards with an initial velocity of 100 m/s in a medium in which the resistance force is equal to 0.01 times the square of the body's velocity, i.e. $0.01v^2$. Use $g = 10 \text{ m/s}^2$.

- (i) Show that the maximum height reached by the particle is $50 \log_e 11$ metres.

4

- (ii) Will the downward velocity of the particle on its return to the point of projection be greater than, less than, or equal to 100 m/s? Justify your answer.

2

- (iii) Calculate the actual downward velocity of the particle on its return to the point of projection.

4

Question 8 (15 marks) Use a SEPARATE sheet of paper.**Marks**

- (a) Use the following identity to answer the following questions.

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}.$$

- (i) Solve $x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$. 3

- (ii) Hence show that

(1) $\tan \frac{\pi}{24} + \tan \frac{7\pi}{24} + 4\sqrt{3} = \tan \frac{5\pi}{24} + \tan \frac{11\pi}{24}$ 1

(2) $\tan \frac{\pi}{24} \tan \frac{5\pi}{24} = \cot \frac{7\pi}{24} \cot \frac{11\pi}{24}$ 1

- (iii) Find the polynomial of least degree that has zeros 3

$$\left(\cot \frac{\pi}{24}\right)^2, \left(\cot \frac{7\pi}{24}\right)^2, \left(\cot \frac{13\pi}{24}\right)^2, \left(\cot \frac{19\pi}{24}\right)^2.$$

- (b) Let $I_n = \int_0^1 x(x^2 - 1)^n dx$ for $n = 0, 1, 2, \dots$

- (i) Use integration by parts to show that 3

$$I_n = \frac{-n}{n+1} I_{n-1} \text{ for } n \geq 1.$$

- (ii) Hence or otherwise show that 2

$$I_n = \frac{(-1)^n}{2(n+1)} \text{ for } n \geq 0.$$

- (iii) Explain why $I_{2n} > I_{2n+1}$ for $n \geq 0$ 1

- (iv) Explain whether or not $I_n > I_{n+2}$ for all $n \geq 0$. 1

End of Examination

2008 Extension 2 Trial Solutions

$$Q1a) \int \frac{dx}{x^2-4x+40} = \int \frac{dx}{(x-2)^2+36}$$

$$= \frac{1}{6} \tan^{-1}\left(\frac{x-2}{6}\right)$$

$$b) \int_0^2 x^3 e^{x^2} dx = \left[\frac{x^2 e^{x^2}}{2} \right]_0^2 - \int_0^2 2x \frac{e^{x^2}}{2} dx \quad u = x^2 \quad v = \frac{1}{2} e^{x^2}$$

$$= \left[\frac{4e^4}{2} - 0 \right] - \frac{1}{2} \left[e^{x^2} \right]_0^2 \quad du = 2x \quad dv = x e^{x^2}$$

$$= 2e^4 - \frac{1}{2}(e^4 - 1)$$

$$= \frac{1}{2}(3e^4 + 1)$$

$$c) \int \sin^3 x dx = \int \sin^2 x \cdot \sin x dx$$

$$= \int (1 - \cos^2 x) \sin x dx$$

$$= \int \sin x - \cos^2 x \sin x dx$$

$$= -\cos x + \frac{\cos^3 x}{3}$$

$$d) \int_0^1 \frac{x}{\sqrt{4-x}} dx \quad \text{let } u = 4-x \Rightarrow x = 4-u$$

$$= \int_4^3 \frac{4-u}{\sqrt{u}} \cdot (-1) du \quad du = -1$$

$$= \int_3^4 (4u^{-1/2} - u^{1/2}) du \quad x=0, u=4$$

$$= \left[8u^{1/2} - \frac{2}{3} u^{3/2} \right]_3^4 \quad x=1, u=3$$

$$= \left(8\sqrt{4} - \frac{2}{3} \cdot 4\sqrt{4} - \left(8\sqrt{3} - \frac{2}{3} \cdot 3\sqrt{3} \right) \right)$$

$$= 16 - \frac{16}{3} - 6\sqrt{3}$$

$$= \frac{1}{3}(32 - 18\sqrt{3})$$

$$Q1i) \frac{3x^2+2x+11}{(x^2+3)(1-x)} = \frac{(ax+b)(1-x)+c(x^2+3)}{(x^2+3)(1-x)}$$

$$ax - ax^2 + b - bx + cx^2 + 3c = 3x^2 + 2x + 11$$

$$c - a = 3 \rightarrow a = c - 3 \quad \text{①}$$

$$a - b = 2 \rightarrow c - 3 - b = 2 \therefore c - b = 5 \quad \text{②}$$

$$3c + b = 11 \quad \text{③}$$

$$\text{①} + \text{②}: 4c = 16, c = 4 \quad \text{ie } a = 1, b = -1, c = 4$$

$$\text{in ② } 4 - b = 5, b = -1$$

$$\text{in ① } a - 4 - 2 = 1$$

$$ii) \int \frac{3x^2+2x+11}{(x^2+3)(1-x)} dx = \int \frac{x-1}{x^2+3} + \frac{4}{1-x} dx$$

$$= \int \frac{1}{2} \cdot \frac{2x}{x^2+3} - \frac{1}{x^2+3} + \frac{4}{1-x} dx$$

$$= \frac{1}{2} \ln(x^2+3) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) - 4 \ln(1-x), x < 1$$

SOLUTIONS QUESTION 2

(a) (i) Given $z = 1 + i$

$$|z| = \sqrt{1^2 + 1^2}$$
$$= \sqrt{2}$$
$$\arg z = \tan^{-1} \frac{1}{1}$$
$$= \frac{\pi}{4}$$

$$\therefore z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \quad (2)$$

(ii) $z^{12} = \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]^{12}$
 $= 2^6 \left(\cos 3\pi + i \sin 3\pi \right)$
 $= 64 (-1 + 0)$
 $= -64$

De Moivre's Thm

(2)

$$(b) \text{ ~~the~~ } P(z) = z^4 - 30z^2 + 289$$

(i) If $p(4+i) = 0$ then $4+i$ is a zero of $p(z)$.

$$\begin{aligned} P(4+i) &= (4+i)^4 - 30(4+i)^2 + 289 && \text{substitution \& partial simp.} \\ &= ((4+i)^2)^2 - 30(15+8i) + 289 \\ &= (15+8i)^2 - 450 - 240i + 289 && \text{--- (1)} \\ &= 225 + 240i + 64i^2 - 450 - 240i + 289 \\ &= 225 + 240i - 64 - 450 - 240i + 289 && \text{--- (1)} \\ &= 0 \end{aligned}$$

$\therefore (4+i)$ is a zero of $P(z)$

(ii) Since $-4+i$ is a zero, so also is $-4-i$ (complex conj.)
 i.e. $[z - (-4+i)][z - (-4-i)]$ is a factor (1)
 $= [z - (-4) - i][z - (-4) + i]$ is a factor
 $= (z+4)^2 - i^2$ " " "
 $= z^2 + 8z + 17$ " " " (1)

(ii) (continued)

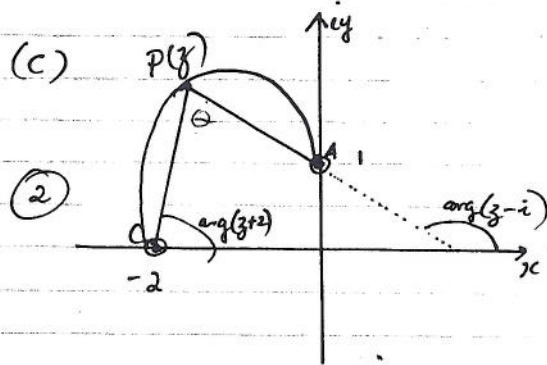
$$\begin{array}{r}
 z^2 + 8z + 17 \\
 \hline
 z^2 - 8z + 17 \quad \bigg) \quad \begin{array}{l} z^4 \\ z^4 - 8z^3 + 17z^2 \\ \hline 8z^3 - 47z^2 \\ 8z^3 - 64z^2 + 136z \\ \hline 17z^2 - 136z + 289 \\ 17z^2 - 136z + 289 \\ \hline 0 \end{array} \quad \begin{array}{l} +289 \\ +289 \end{array} \\
 \hline
 \end{array}
 \quad (1)$$

$\therefore z^2 + 8z + 17$ is a factor of $P(z)$

$$\begin{aligned} \therefore z^2 + 8z + 17 &= (z^2 + 8z + 16) + 1 \quad (\text{Alt: use quad. form}) \\ &= (z+4)^2 - i^2 \\ &= (z+4-i)(z+4+i) \\ &= (z-(-4+i))(z-(-4-i)) \quad (1) \end{aligned}$$

$\therefore$ remaining zero's are $-4+i, -4-i$

$\therefore$ zeros of $P(z)$ are $4+i, 4-i, -4+i, -4-i$. (1)



$\therefore P(z)$ is a semicircle with AC as diameter and P above AC. Points A, C are excluded from the locus. $\odot 2$

Geometrically
 $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{2}$

$$\arg(z-i) - \arg(z+2) = \frac{\pi}{2}$$

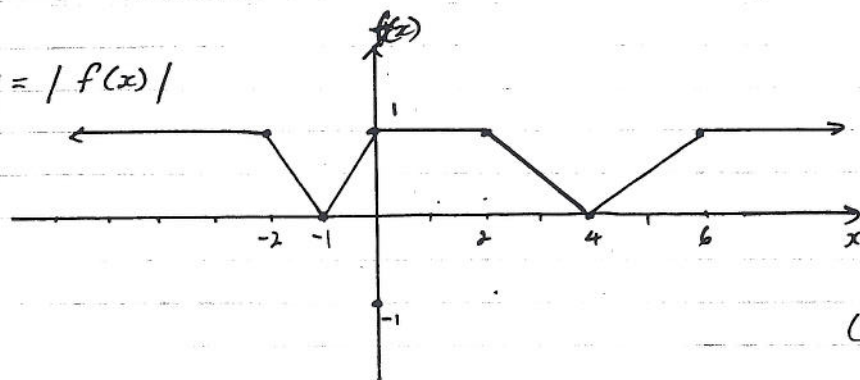
4 NB $\arg(z-i) > \arg(z+i)$

From diagram
Now $\theta + \arg(z+2) = \arg(z-i)$
(ext. angle of Δ)
 $\therefore \theta = \arg(z-i) - \arg(z+2)$
 $= \frac{\pi}{2}$

$$\therefore Q = \arg(z-i) - \arg(z+2) = \frac{\pi}{2}$$

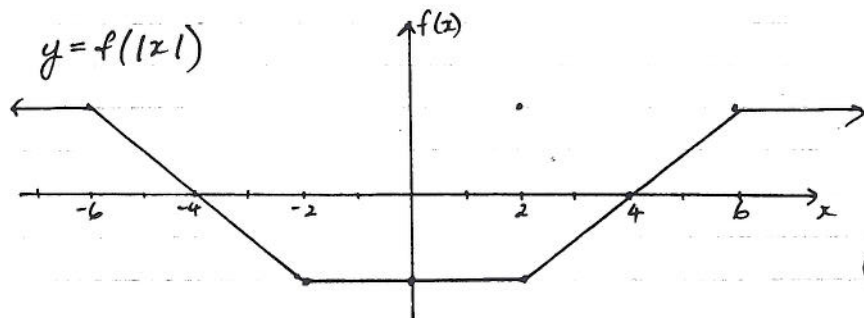
SOLUTIONS QUESTION 3

(a) (i) $y = |f(x)|$



(1)

(ii) $y = f(|x|)$



(1)

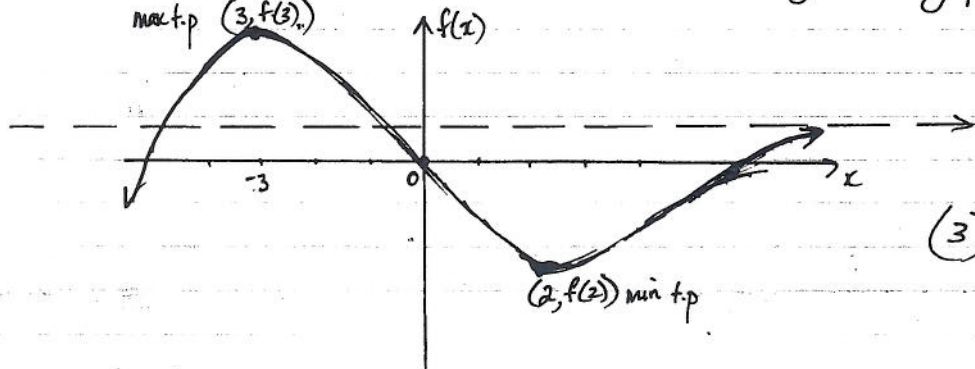
(b) $f'(2) = 0$ $f'(2^-) < 0$ $f'(2^+) > 0$ $\therefore x = 2$ is min
 $f'(-3) = 0$ $f'(-3^-) > 0$ $f'(-3^+) < 0$ $\therefore x = -3$ is max

Curve passes through (0, 0)

Curve below x axis when $x = 1$

as $x \rightarrow \infty$, $f'(x) \rightarrow 0$ ie flattens out to horizontal asymptote.

max t.p. (3, f(3))



(3)

(c) $y = \frac{x^3 + 4}{x^2}$

(i) $= x + 4x^{-2}$

T.P.s. ($\frac{dy}{dx} = 0$)

$\frac{dy}{dx} = 1 - \frac{8}{x^3}$

$\frac{dy}{dx} = 0$ when $x = 2$

zero's $x^3 = -4$
 $x = \sqrt[3]{-4}$

Inflections ($\frac{d^2y}{dx^2} = 0$)

$\frac{d^2y}{dx^2} = \frac{24}{x^4} > 0 \therefore$ no inflexions

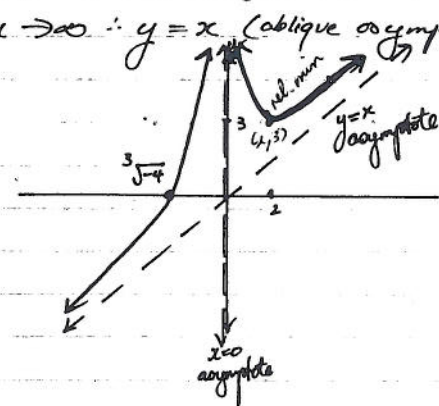
test t.p.

$\frac{d^2y}{dx^2} > 0$ when $x = 2 \therefore$ min t.p. at (2, 3)

Asymptotic Behaviour
vertical $x = 0$

$x \rightarrow 0^+ y \rightarrow \infty$
 $x \rightarrow 0^- y \rightarrow -\infty$

other as $x \rightarrow \infty \therefore y = x$ (oblique asymptote)



(3)

(ii) From the graph $x^3 - kx^2 + 4 = 0 \Rightarrow k = \frac{x^3 + 4}{x^2}$ what horizontal line $y = k$ will only have one solution? any $k < 3 \cdot x^2$ (1)

(d) (i) If 'a' is a mult. root of $P(x)$ then

$$P(x) = (x-a)^r \cdot Q(x)$$

$$P'(x) = (x-a)^r \cdot Q'(x) + Q(x) \cdot r(x-a)^{r-1}$$

$$= (x-a)^{r-1} [(x-a)Q'(x) + rQ(x)]$$

$$= (x-a)^{r-1} S(a) \quad \text{where } S(a) = (x-a)Q'(x) + rQ(x)$$

$$\therefore P'(a) = (a-a)^{r-1} S(a)$$

$$= 0 \times S(a)$$

$$= 0 \quad \text{as required.}$$

(3)

(ii) $P(x) = 16x^3 - 12x^2 + 1$

$$P'(x) = 48x^2 - 24x$$

when $P'(x) = 0$

$$48x^2 - 24x = 0$$

$$24x(2x-1) = 0$$

$$\therefore x = 0, \frac{1}{2}$$

Since $P(0) = 1$

and $P(\frac{1}{2}) = 0$ $x = \frac{1}{2}$ is double root

Now

$$\alpha + \beta + \gamma = \frac{12}{16}$$

but $\alpha = \beta = \frac{1}{2}$

$$\frac{1}{2} + \frac{1}{2} + \gamma = \frac{12}{16}$$

$$\gamma = -\frac{1}{4}$$

$\therefore$ 3 roots are $\frac{1}{2}, \frac{1}{2}, -\frac{1}{4}$

(3)

Q4.a) $x = \sqrt{2} \cos \theta$

$y = 3 \sin \theta$

$$\cos \theta = \frac{x}{\sqrt{2}}$$

$$\sin \theta = \frac{y}{3}$$

$$\left(\frac{x}{\sqrt{2}}\right)^2 + \left(\frac{y}{3}\right)^2 = 1 \quad (\cos^2 \theta + \sin^2 \theta = 1)$$

$$\therefore \frac{x^2}{2} + \frac{y^2}{9} = 1$$

$$a^2 = b^2(1-e^2), \quad a < b$$

$$2 = 9(1-e^2)$$

$$1-e^2 = \frac{2}{9}$$

$$e^2 = \frac{7}{9}$$

$$\therefore e = \frac{\sqrt{7}}{3}$$

b) i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

at $P(a \cos \theta, b \sin \theta)$, $\frac{dy}{dx} = -\frac{b^2 \cdot a \cos \theta}{a^2 b \sin \theta}$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

$\therefore$ equation of tangent is $y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

$$a \sin \theta y - ab \sin^2 \theta = -b \cos \theta x + ab \cos^2 \theta$$

$$b \cos \theta x + a \sin \theta y = ab(\sin^2 \theta + \cos^2 \theta)$$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad \text{--- (1)}$$

ii) at T, $y = 0$ & T lies on tangent $\therefore \frac{x \cos \theta}{a} = 1$

$$\text{ie } x = \frac{a}{\cos \theta}$$

$$\therefore T\left(\frac{a}{\cos \theta}, 0\right)$$

equation of directrix is $x = \frac{a}{e}$

$$\therefore \frac{a}{\cos \theta} = \frac{a}{e}$$

$$\text{ie } \cos \theta = e$$

- 4b) iii) since $\cos \theta = e$, the x-coordinate of P is ae
 The focus has coordinates $S(ae, 0)$
 $\therefore$ The focal chord makes an angle of 90° with x-axis

iv) $\Delta TOR \parallel \Delta TSP$ (equiangular)

$$\therefore \frac{RT}{OT} = \frac{RP}{OS} \text{ (ratio of intercepts)}$$

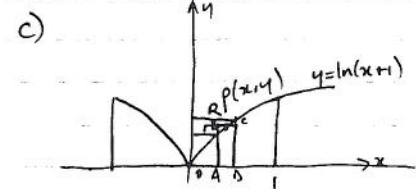
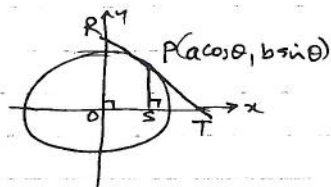
$$T\left(\frac{a}{e}, 0\right) \text{ and } S(ae, 0)$$

$$\therefore OT = \frac{a}{e} \text{ and } OS = ae$$

$$\therefore \frac{RT}{\frac{a}{e}} = \frac{RP}{ae}$$

$$RT = \frac{RP}{ae} \cdot \frac{a}{e}$$

$$\therefore RP = e^2 RT$$



let ABCD be rectangle of height y
 and width δx where $P(x, y)$ is
 the midpt of the rectangle

let R be outer radius & r be inner radius

$$\delta V = \pi(R^2 - r^2) \times h$$

$$= \pi(R+r)(R-r) \times y$$

$$= 2\pi\left(\frac{R+r}{2}\right)(R-r) \times y$$

$$= 2\pi \cdot x \cdot \delta x \cdot y$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi xy \delta x, \quad y = \ln(x+1)$$

$$= \int_0^1 2\pi x \cdot \ln(x+1) dx$$

$$u = \ln(x+1) \quad v = \frac{x^2}{2}$$

$$du = \frac{1}{x+1} \quad dv = x$$

$$= 2\pi \left[\frac{x^2}{2} \ln(x+1) \right]_0^1 - \int_0^1 \frac{x^2}{2(x+1)} dx$$

$$= 2\pi \left(\frac{1}{2} \ln 2 - \frac{1}{2} \int_0^1 \left(x-1 + \frac{1}{x+1} \right) dx \right)$$

$$= 2\pi \left(\frac{1}{2} \ln 2 - \frac{1}{2} \left[\frac{x^2}{2} - x + \ln(x+1) \right]_0^1 \right)$$

$$= 2\pi \left(\frac{1}{2} \ln 2 - \frac{1}{2} \left[\frac{1}{2} - 1 + \ln 2 - (0-0+0) \right] \right)$$

$$\therefore \text{Volume} = \frac{\pi}{2} \text{ units}^3$$

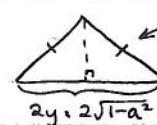
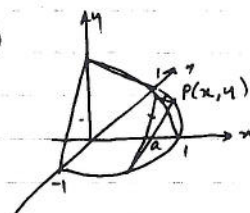
$$Q5 a) i) \int_{-a}^a \sqrt{a^2 - x^2} dx = \frac{1}{2} \pi \cdot a^2, \text{ area of semicircle centre } (0,0), \text{ radius } a$$

$$= \frac{\pi a^2}{2}$$

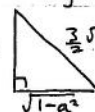
$$ii) f(x) = x \sqrt{a^2 - x^2} \text{ is an odd fn } \therefore \int_{-a}^0 f(x) dx = - \int_0^a f(x) dx$$

$$\therefore \int_{-a}^a f(x) dx = 0$$

b) i)



$$\frac{3}{4} \times 2\sqrt{1-a^2} = \frac{3\sqrt{1-a^2}}{2}$$



$$\text{height} = \sqrt{\frac{9}{4}(1-a^2) + (1-a^2)}$$

$$= \sqrt{\frac{5}{4}(1-a^2)}$$

$$= \frac{\sqrt{5}}{2} \sqrt{1-a^2}$$

$$\therefore \text{Area of } \Delta = \frac{1}{2} \times 2\sqrt{1-a^2} \times \frac{\sqrt{5}}{2} \sqrt{1-a^2}$$

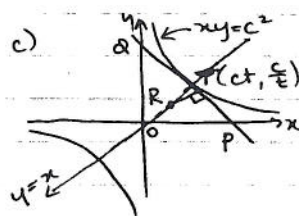
$$= \frac{\sqrt{5}}{2} (1-a^2)$$

$$ii) V = \int_0^1 \frac{\sqrt{5}}{2} (1-x^2) dx$$

$$= \frac{\sqrt{5}}{2} \left[x - \frac{x^3}{3} \right]_0^1$$

$$= \frac{\sqrt{5}}{2} \left(1 - \frac{1}{3} \right)$$

$$\therefore \text{Vol} = \frac{\sqrt{5}}{3} \text{ units}^3$$



$$i) xy = c^2 \Rightarrow y = c^2 x^{-1}$$

$$\frac{dy}{dx} = -c^2 x^{-2}$$

$$\text{at } T(ct, \frac{c}{t}) \quad MT = \frac{-c^2}{(ct)^2} = -\frac{1}{t^2}$$

$$\therefore \text{eqn of tangent is } y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$$

$$t^2 y - ct = -x + ct$$

$$x + t^2 y = 2ct$$

$$ii) \text{at } P, y=0 \therefore x=2ct \therefore P(2ct, 0)$$

$$\text{at } Q, x=0 \therefore t^2 y = 2ct \therefore Q(0, \frac{2c}{t})$$

$$iii) MN = t^2 \therefore \text{eqn is } y - \frac{c}{t} = t^2 (x - ct)$$

$$\text{ie } ty - c = t^3 x - ct^4$$

$$t^3 x - ty = ct^4 - c$$

Question 6 Solutions

iv) R lies on $y=x$ and the normal at T, $t^3x - ty = ct^4 - c$

$$\therefore t^3x - tx = c(t^4 - 1)$$

$$tx(t^2 - 1) = c(t^2 - 1)(t^2 + 1)$$

$$x = \frac{c}{t}(t^2 + 1)$$

v) R lies on $y=x$ $\therefore$ coordinates of R $(\frac{c}{t}(t^2 + 1), \frac{c}{t}(t^2 + 1))$

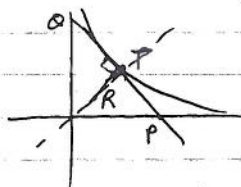
$$P(2ct, 0) \quad Q(0, \frac{2c}{t})$$

$$\text{Midpt of PQ is } (\frac{2ct+0}{2}, \frac{0+\frac{2c}{t}}{2})$$

$$= (ct, \frac{c}{t}), \text{ which is the point T.}$$

$\therefore$ Any point on the normal at T will lie on the perpendicular bisector of PQ.

$\therefore \triangle PRQ$ is isosceles



(a) (i) let $x = -X$

$$(-X)^3 + 2(-X) - 1 = 0$$

$$-X^3 - 2X - 1 = 0$$

$$\therefore X^3 + 2X + 1 = 0 \quad (1)$$

(ii) let $x = \sqrt{X}$

$$(\sqrt{X})^3 + 2\sqrt{X} - 1 = 0$$

$$X\sqrt{X} + 2\sqrt{X} = 1 \quad (\text{sq. b.s.})$$

$$X^3 + 4X^2 + 4X = 1$$

$$\therefore X^3 + 4X^2 + 4X - 1 = 0 \quad (2)$$

(iii) let $x = \pm X$ (Note $x = -X$) in part (i) and $x = X$ is $\therefore (X^3 + 2X - 1)(X^3 + 2X + 1) = 0$ must have $x = \pm X$ original equation

$$\therefore (X^3 + 2X - 1)(X^3 + 2X + 1) = 0$$

$$X^6 + 4X^4 + 4X^2 - 1 = 0 \quad (2)$$

(b) Note $1-i$ is NOT a root of $x^2 + (a+2i)x + 5-ib = 0$ since all coefficients are NOT real

$1+i$ is a root of $x^2 + (a+2i)x + 5-ib = 0$ (given)

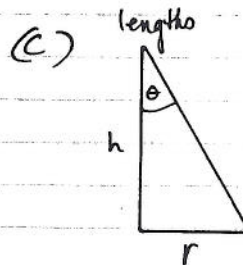
$$\therefore (1+i)^2 + (a+2i)(1+i) + 5-ib = 0$$

$$\therefore 1+2i-1+a+2i+ai-2+5-ib = 0$$

$$(a+3) + (4+a-b)i = 0$$

$$\therefore a+3=0 \quad \text{as } \text{Re} = 0 \Rightarrow a = -3$$

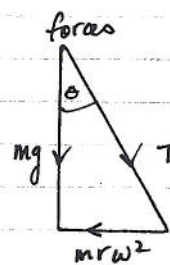
$$4+a-b=0 \quad \text{as } \text{Im} = 0 \Rightarrow b = 1 \quad (3)$$



$$r = 0.2 \sin \theta$$

$$= \frac{\sin \theta}{5}$$

$$\omega = \frac{72 \times 2\pi}{60} = \frac{12\pi}{5} \text{ rad/s}$$



$$(i) T \sin \theta = mrw^2 \quad (\text{hor.})$$

$$T \cos \theta = mg \quad (\text{vert.})$$

$$\therefore \tan \theta = \frac{rw^2}{g}$$

$$\therefore \tan \theta = \frac{\sin \theta}{5} \times \frac{144\pi^2}{25} \times \frac{1}{10}$$

$$= \frac{72\pi^2 \sin \theta}{625} \quad \text{as req.}$$

(3)

Question 7 Solutions

(ii) $\tan \theta = \frac{72\pi^2 \sin \theta}{625}$ from (i)

$$\frac{\sin \theta}{\cos \theta} = \frac{72\pi^2 \sin \theta}{625}$$

$$\therefore \cos \theta = \frac{625}{72\pi^2}$$

$$\theta = 28^\circ 25'$$

(2)

(iii) from (i) $\tan \theta = \frac{r\omega^2}{g}$

$$\frac{r}{h} = \frac{r\omega^2}{g}$$

$$h = \frac{g}{\omega^2}$$

as can be seen h depends

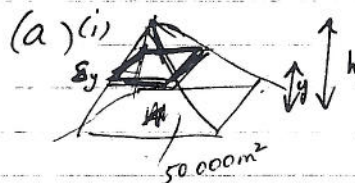
on g a constant and

ω which is the same as

before $\therefore h$ remains the same

$\therefore$ double mass $\Rightarrow$ no effect on θ .

(2)



Since a square pyramid
If $A_1 : A_2$ then $h_1^2 : h_2^2$

$$\frac{A(y)}{A} = \frac{(h-y)^2}{h^2} \quad \text{similar areas (all squares are similar)}$$

$$\therefore A(y) = \frac{50 \times 10^4}{h^2} \times (h-y)^2 \quad (*)$$

(ii) Volume = $\sum_{y=0}^h A(y) \delta y$ ✓

$$= \int_0^h \frac{A(h-y)^2}{h^2} dy$$

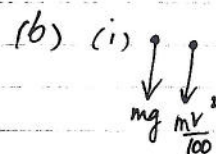
$$= \left[-\frac{A}{3h^2} (h-y)^3 \right]_0^h \quad \checkmark$$

$$= -\frac{A}{3h^2} (0 - h^3)$$

$$= \frac{Ah}{3} \quad \checkmark$$

$$= \frac{50\,000 \times 150}{3}$$

$$= 2\,500\,000 \text{ m}^3 \quad \checkmark$$



note $m=1\text{kg}$

$$m\ddot{x} = -10m - \frac{mv^2}{100} \quad (m=1\text{kg})$$

$$\therefore \ddot{x} = -\left(10 + \frac{v^2}{100}\right)$$

$$v \frac{dv}{dx} = -\left(\frac{1000 + v^2}{100}\right)$$

$$\frac{dv}{dx} = -\left(\frac{1000 + v^2}{100v}\right)$$

$$\therefore \frac{dx}{dv} = -\frac{100v}{1000 + v^2}$$

$$\therefore x = -50 \ln(1000 + v^2)$$

when $x=0$ $v=100$

$$\therefore C = 50 \ln 11000$$

$$\therefore x = 50 \ln \frac{11000}{1000 + v^2}$$

when $v=0$ (highest pt)

$$x = 50 \ln 11 \quad (4)$$

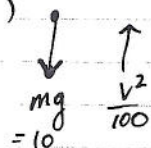
(ii) The downward velocity will be less than 100 m/s (its original). For example consider a spear gun being fired upwards in water from low in the ocean then after reaching its highest point, slowly drifting back downwards.

or more everyday

A football being kicked from a foot directly upwards high velocity at impact, highest point zero velocity. person is able to catch it on return (lower velocity at return, otherwise unable to catch it.)

(2)

(iii)



$$m\ddot{x} = 10 - \frac{v^2}{100} \quad (m=1)$$

$$\ddot{x} = \frac{1000 - v^2}{100}$$

$$v \frac{dv}{dx} = \frac{1000 - v^2}{100}$$

$$\frac{dv}{dx} = \frac{1000 - v^2}{100v}$$

$$\frac{dx}{dv} = \frac{100v}{1000 - v^2}$$

$$x = -50 \ln(1000 - v^2) + c \quad (\text{at top } x=0, v=0)$$

$$\therefore C = 50 \ln 1000$$

$$x = 50 \ln \frac{1000}{1000 - v^2}$$

$$11 = \frac{1000}{1000 - v^2}$$

$$11000 = 11000 - 11v^2 = 1000$$

$$10000 = 11v^2$$

$$v^2 = \frac{10000}{11} \quad v = \frac{100}{\sqrt{11}} \text{ m/s} = \frac{100\sqrt{11}}{11} \text{ m/s.} \quad (4)$$

when $x = 50 \ln 11$
(distance of return)
from highest pt

Q2.a)i)

$$\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$$

$$\text{let } x = \tan\theta \quad \text{and} \quad \tan 4\theta = \frac{1}{\sqrt{3}}$$

$$\therefore \frac{4x - 4x^3}{1 - 6x^2 + x^4} = \frac{1}{\sqrt{3}}$$

$$\therefore x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$$

$$\tan 4\theta = \frac{1}{\sqrt{3}}$$

$$4\theta = n\pi + \frac{\pi}{6}$$

$$\theta = \frac{n\pi}{4} + \frac{\pi}{24}, \quad n=0,1,2,3$$

$$n=0, \quad \theta = \frac{\pi}{24} \quad \text{so } x = \tan \frac{\pi}{24}$$

$$n=1, \quad \theta = \frac{\pi}{4} + \frac{\pi}{24} = \frac{7\pi}{24}, \quad x = \tan \frac{7\pi}{24}$$

$$n=2, \quad \theta = \frac{\pi}{2} + \frac{\pi}{24} = \frac{13\pi}{24}, \quad x = \tan \frac{13\pi}{24} \quad (\text{or } -\tan \frac{11\pi}{24})$$

$$n=3, \quad \theta = \frac{3\pi}{4} + \frac{\pi}{24} = \frac{19\pi}{24}, \quad x = \tan \frac{19\pi}{24} \quad (\text{or } -\tan \frac{5\pi}{24})$$

ii) (1) from (i) $\tan \frac{\pi}{24} + \tan \frac{7\pi}{24} - \tan \frac{5\pi}{24} - \tan \frac{11\pi}{24} = -4\sqrt{3}$ (sum of roots)

$$\therefore \tan \frac{\pi}{24} + \tan \frac{7\pi}{24} + 4\sqrt{3} = \tan \frac{5\pi}{24} + \tan \frac{11\pi}{24}$$

ii) (2) $(\tan \frac{\pi}{24})(\tan \frac{7\pi}{24})(-\tan \frac{5\pi}{24})(-\tan \frac{11\pi}{24}) = 1$ (product of roots)

$$\tan \frac{\pi}{24} \tan \frac{5\pi}{24} = \frac{1}{\tan \frac{7\pi}{24} \tan \frac{11\pi}{24}}$$

$$\therefore \tan \frac{\pi}{24} \tan \frac{5\pi}{24} = \cot \frac{7\pi}{24} \cot \frac{11\pi}{24}$$

iii) let $\alpha = \tan \frac{\pi}{24}$

$$\frac{1}{\alpha^2} = (\cot \frac{\pi}{24})^2$$

$$\text{let } \frac{1}{\alpha^2} = x \quad \text{so } \alpha = \frac{1}{\sqrt{x}}$$

$$\left(\frac{1}{\sqrt{x}}\right)^4 + 4\sqrt{3}\left(\frac{1}{\sqrt{x}}\right)^3 - 6\left(\frac{1}{\sqrt{x}}\right)^2 - 4\sqrt{3}\left(\frac{1}{\sqrt{x}}\right) + 1 = 0$$

$$\frac{1}{x^2} + \frac{4\sqrt{3}}{x\sqrt{x}} - \frac{6}{x} - \frac{4\sqrt{3}}{\sqrt{x}} + 1 = 0$$

$$x^2 - 4\sqrt{3}x\sqrt{x} - 6x + 4\sqrt{3}\sqrt{x} + 1 = 0$$

$$4\sqrt{3}\sqrt{x}(x-1) = x^2 - 6x + 1$$

$$48x(x-1)^2 = (x^2 - 6x + 1)^2$$

$$48x^3 - 96x^2 + 48x = x^4 - 12x^3 + 2x^2 + 36x^2 - 12x + 1$$

$$\text{ie } x^4 - 60x^3 + 134x^2 - 60x + 1 = 0$$

8b) i) $I_n = \int_0^1 x(x^2-1)^n dx$

$$= \left[\frac{x^2}{2} (x^2-1)^n \right]_0^1 - \int_0^1 \frac{x^2}{2} \cdot n(x^2-1)^{n-1} \cdot 2x dx$$

$$= \left[\frac{1}{2} \times 0 - 0 \right] - n \int_0^1 x^3 (x^2-1)^{n-1} dx$$

$$= -n \int_0^1 \frac{x^3 (x^2-1)^n}{x^2-1} dx$$

$$= -n \int_0^1 \left(x + \frac{x}{x^2-1} \right) (x^2-1)^n dx$$

$$= -n \left[\int_0^1 x(x^2-1)^n dx + \int_0^1 \frac{x}{x^2-1} (x^2-1)^n dx \right]$$

$$= -n \int_0^1 x(x^2-1)^n dx - n \int_0^1 x(x^2-1)^{n-1} dx$$

$$\therefore I_n = -n I_n - n I_{n-1}$$

$$(1+n) I_n = -n I_{n-1}$$

$$I_n = \frac{-n}{(n+1)} I_{n-1} \quad \text{for } n \geq 1$$

ii) "Hence": $I_n = \frac{-n}{n+1} I_{n-1}$

$$= \frac{-n}{n+1} \cdot \frac{-n+1}{n} \cdot \frac{-n+2}{n-1} \cdots \frac{-3}{4} \cdot \frac{-2}{3} \cdot \frac{-1}{2} I_0$$

$$I_0 = \int_0^1 x(x^2-1)^0 dx$$

$$= \left[\frac{x^2}{2} \right]_0^1$$

$$= \frac{1}{2}$$

$$\text{So } I_n = (-1)^n \frac{n}{n+1} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n-1} \cdots \frac{3}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{2}$$

$$= \frac{(-1)^n}{2(n+1)}, \quad n \geq 0$$

"Otherwise": $I_n = \int_0^1 x(x^2-1)^n dx$

$$= \frac{1}{2} \int_0^1 2x(x^2-1)^n dx$$

$$= \frac{1}{2} \left[\frac{(x^2-1)^{n+1}}{n+1} \right]_0^1$$

$$= \frac{1}{2(n+1)} (0 - (-1)^{n+1})$$

$$= \frac{(-1)^n}{2(n+1)}, \quad n \geq 0$$

iii) $I_0 = \frac{1}{2}, I_1 = -\frac{1}{4}, I_2 = \frac{1}{6}, I_3 = -\frac{1}{8}, I_4 = \frac{1}{10}, I_5 = -\frac{1}{12}$

ie $I_{2n} > 0$ and $I_{2n+1} < 0$

so $I_{2n} > I_{2n+1}$

OR: $I_{2n} = \frac{(-1)^{2n}}{2(2n+1)}$

$$= \frac{1}{2(2n+1)}$$

$$> 0$$

$$I_{2n+1} = \frac{(-1)^{2n+1}}{2((2n+1)+1)}$$

$$= \frac{-1}{4(n+1)}$$

$$< 0$$

so $I_{2n} > I_{2n+1}$

iv) from (iii), if n is even, $I_n > I_{n+2}$ ie $I_2 = \frac{1}{6} > I_4 = \frac{1}{10} > I_6 = \frac{1}{14}$

if n is odd, $I_1 = -\frac{1}{4} < I_3 = -\frac{1}{8} < I_5 = -\frac{1}{12}$

$\therefore I_n \neq I_{n+2}$ for all $n \geq 0$